



Field Analysis Report

1. Vegetation Indices

Index	Value	Description	Status
NDVI	0.1515	Vegetation health	Low
EVI	0.1152	Enhanced vegetation	Low
NDWI	-0.2133	Water content	Dry
NDRE	0.0801	Red-edge / early stress	Monitor
LAI	0.4300	Leaf area index	Low

2. Crop Health Assessment

Prediction: Diseased (Confidence: 59.6%)

Class	Probability	Bar
Diseased	59.6%	██
Healthy	8.3%	██
Stressed	32.1%	██

3. Irrigation Recommendation

Recommendation	Irrigate Soon
Soil Moisture	23.2%
Water Needed	18.4 mm
Confidence	90%

4. Yield Prediction

Predicted Yield	4.617 tons/hectare
Lower Bound (95%)	3.692 t/ha
Upper Bound (95%)	5.541 t/ha
Harvest Readiness	100.0%

5. Soil Condition Assessment

Soil pH	7.72 (Alkaline)
Salinity	2.218 dS/m (Moderate)
Organic Matter	1.009% (Low — add compost)

6. Variable Rate Application (VRA)

Fertility Zone	Low
Fertiliser Recommendation	Apply 120 kg/ha NPK — soil needs full nutrition
Confidence	96%

This report was generated automatically by AgroSense — a satellite-based AI crop monitoring system developed at Sindh Madressatul Islam University (SMIU) as a Final Year Project (2025–2026). Results are based on Sentinel-2 satellite imagery processed via Google Earth Engine and machine learning models. This report is for advisory purposes only.